RESOLVE · CVPR 2027 · MECHANICAL CAD EDITING

ReSolve CAD Update

Work from 7–9 October 2026 Status as of 9 Oct, 18:30 IST Branch `cvpr2027-research-package`

Over three days we rebuilt the paper's Section 5.2 around the agentic pipeline, ran it with free planners, tested five ways to improve Astra, added two papers' metrics and the Parametric CAD Bench v2/v3 datasets, and started a 200-task CAD Bench run on Kaggle.

PIPELINE, OPEN PLANNERS

+13 tasks, both planners

Qwen3.5-35B 53 → 66 (p = 0.0023); Gemini Flash-Lite 22 → 35 (p = 0.001), on 124 held-out BenchCAD edits.

IMPROVING ASTRA

Still 89 / 124

Pipeline, DPO checker, two FEM checkers and a semantic verifier all fail to beat Astra alone. Perfect selection among its drafts would give 98.

EVALUATION

2 new benchmark versions

CAD Bench v2 reproduces the leaderboard exactly; the v3 verifier scores 100/100 on reference answers and 0 on wrong ones.

"Strict" below means volume IoU ≥ 0.99999 against the reference, scored locally. Paired comparisons use exact McNemar tests; "+15/−2" means 15 tasks gained and 2 lost.

1. Architecture and pipeline

The system has one job: take a source CAD program and an edit instruction, and return the edited program. A frozen planner proposes edits; the agentic pipeline checks, repairs and selects them without any reference answer. References are used only for scoring.

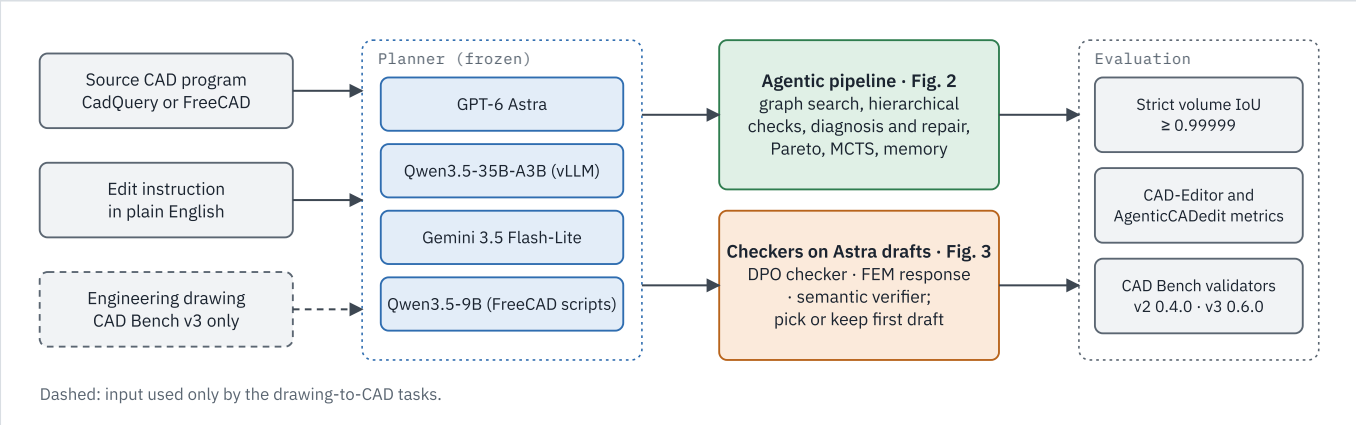


Figure 1. System overview. Every planner is used as-is, with no fine-tuning. The pipeline (Fig. 2) wraps any planner; the checkers (Fig. 3) were tested only on Astra's saved drafts. Evaluation is the only place reference shapes are used.

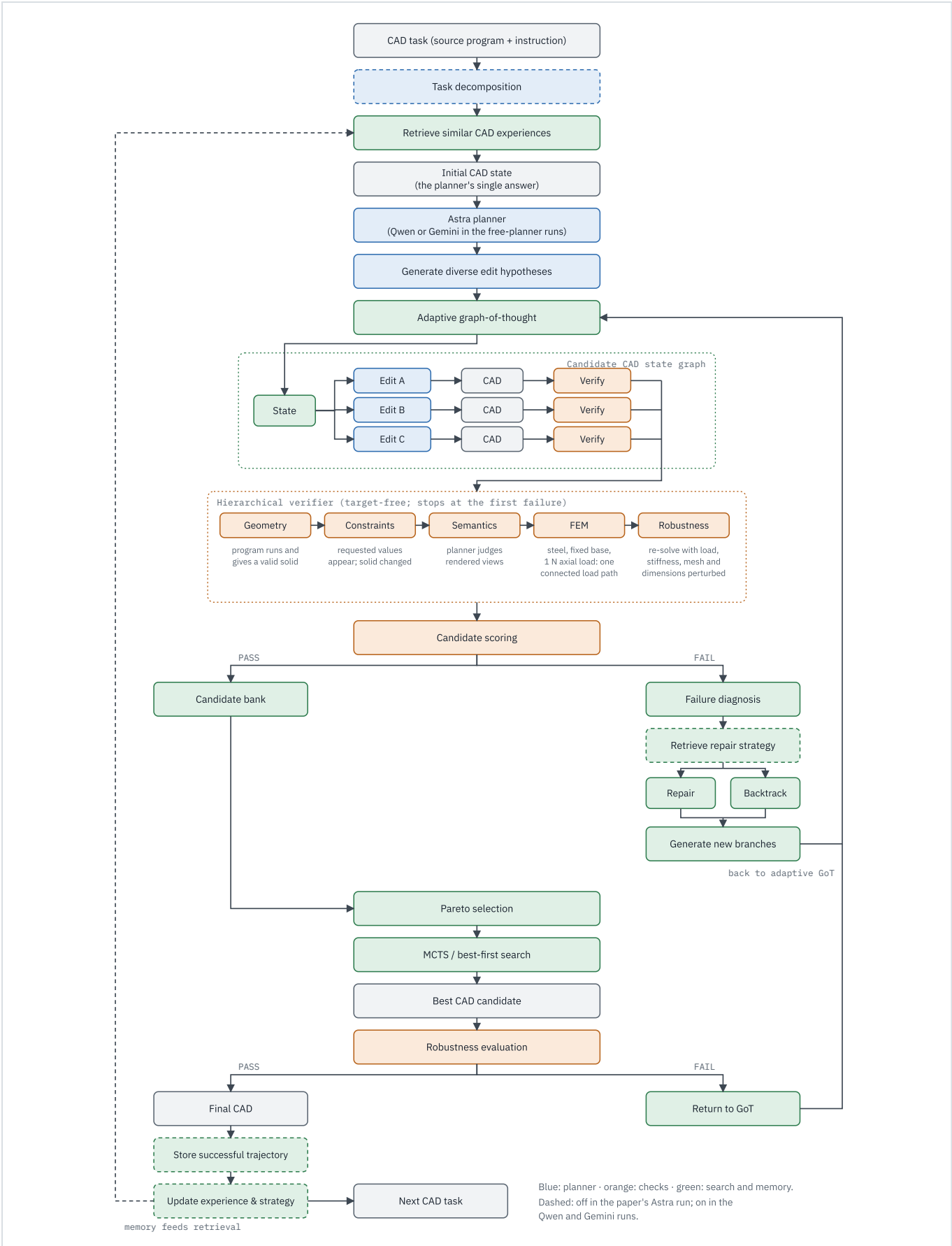


Figure 2. Agentic CAD pipeline, drawn box for box from the pipeline design. The planner's single answer is the initial CAD state. Each hypothesis becomes an edit, an executed CAD model and a verification in the candidate state graph. The hierarchical verifier runs five target-free stages and stops at the first failure. **FEM** is the fourth stage: a linear-elastic tetrahedral solve (steel, E = 210 GPa, v = 0.3; lower 5% band fixed, 1 N axial load on the upper 5%; mesh size = part size / 18). It runs on the 115 of 124 tasks whose source part meshes and solves, and passes when the edited part still carries the load as one connected solid. It is a physical sanity gate, not a comparison with the correct edit's response; the robustness stage repeats it

with load, stiffness, mesh and dimensions perturbed. The continuous expected-response FEM checker is separate (Fig. 3, section 5). Failed states are diagnosed and repaired or backtracked into new branches; passing states go to the candidate bank, Pareto selection and MCTS. Memory starts with the 492 BenchCAD training edits; in the Qwen and Gemini runs, held-out tasks that passed every check were added (74 and 49).

2. Paper: Section 5.2 rebuilt around the pipeline

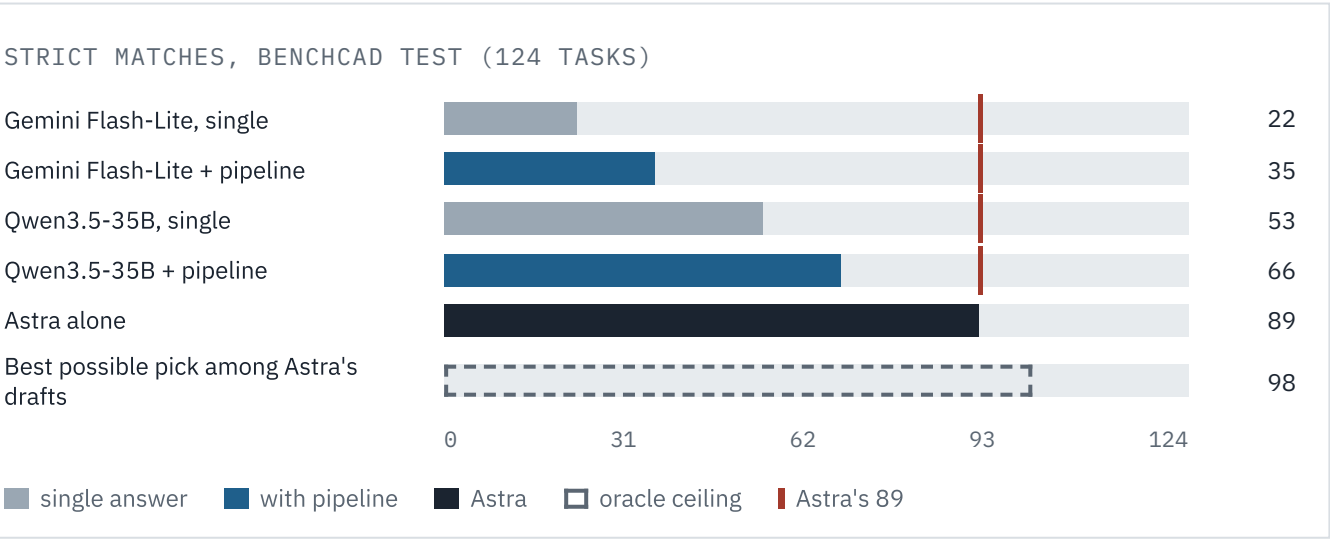
- **Removed:** everything about the Qwen student: distillation, plan anchoring, best-of-n, GRPO, student ReAct and difficulty tiers. The original 3B patch adapter (39/252) stays as the base model in Section 5.1.
- **Added:** `sec:pipeline`, the Astra frontier comparison plus the agentic pipeline diagram as a TikZ figure. The pipeline runs task decomposition, experience retrieval, the Astra planner, adaptive graph-of-thoughts, a hierarchical verifier (geometry → constraints → semantics → FEM → robustness), diagnosis and repair, Pareto selection, MCTS, a final robustness check, and stores each trajectory in memory.
- **Also updated:** the abstract, the conclusion, a new appendix protocol (`sec:pipeline-protocol`) and the Overleaf copy. Macros come from `scripts/pipeline_paper_results.py`.

ASTRA + PIPELINE VARIANT (BENCHCAD TEST)	STRICT / 124	VS ASTRA ALONE
Astra alone	89	baseline
Graph-of-thoughts + target-free checks	89	NO CHANGE
+ MCTS, Pareto, robustness	88	+1 / −2
Full pipeline, with view semantics	85	+2 / −6, P = 0.29
Retrieval-only answer (chosen after seeing results)	94	+7 / −2, P = 0.18

The retrieval-only variant was picked post hoc and is not significant, so the paper does not claim it as a gain.

3. The pipeline lifts open planners significantly

Astra's API credit ran out after one live task, so we ran the same pipeline with free planners: Qwen3.5-35B-A3B served by vLLM on an H100, and Gemini 3.5 Flash-Lite on the free tier. Each planner's own single answer is its paired baseline, and the planner also judges the rendered views. No Astra is involved, and references are used only for the final score.



PLANNER	SINGLE	PIPELINE	GAINED / LOST	MCNEMAR P
Qwen3.5-35B-A3B (self-hosted, free)	53	66	+15 / −2	0.0023
Gemini 3.5 Flash-Lite (free tier, 766 calls)	22	35	+14 / −1	0.001

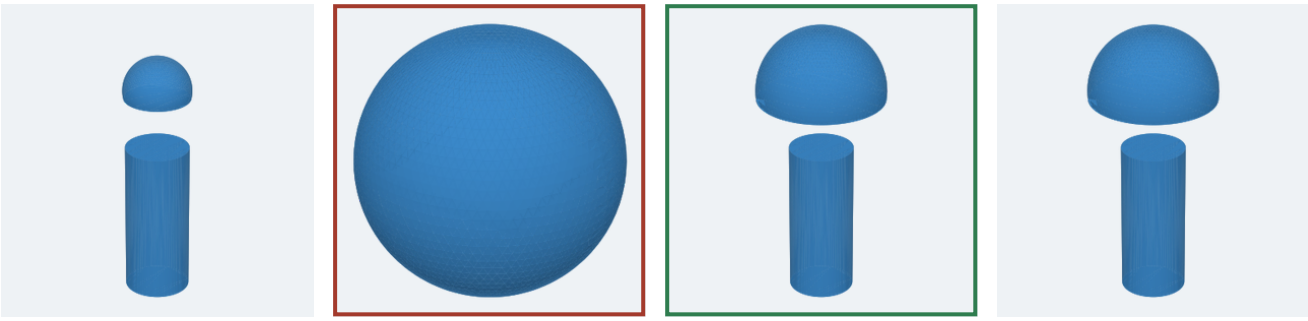
Why it helps weak planners and not Astra: weak planners mostly fail with invalid or non-executing edits, which the diagnosis and repair loop can see and fix. Astra's misses mostly build and pass every check (section 6), so the loop has nothing to act on.

4. Examples of CAD edits

Real BenchCAD test tasks, rendered from the programs each system wrote. Within each example, panels share one scale and view. A green border marks an exact match with the reference (volume IoU ≥ 0.99999); a red border marks a miss.

FIXED BY THE PIPELINE **Rivet: head radius** `BenchCAD · tier T3 · 9b5dda46`

> *"Set the head sphere radius equal to twice the shank radius; the shank, length, and chamfer cut stay the same."*



Source
head radius 3.2

Qwen, single answer
IoU 0.004 · shown at its own scale, about 7× larger

Qwen + pipeline
exact match

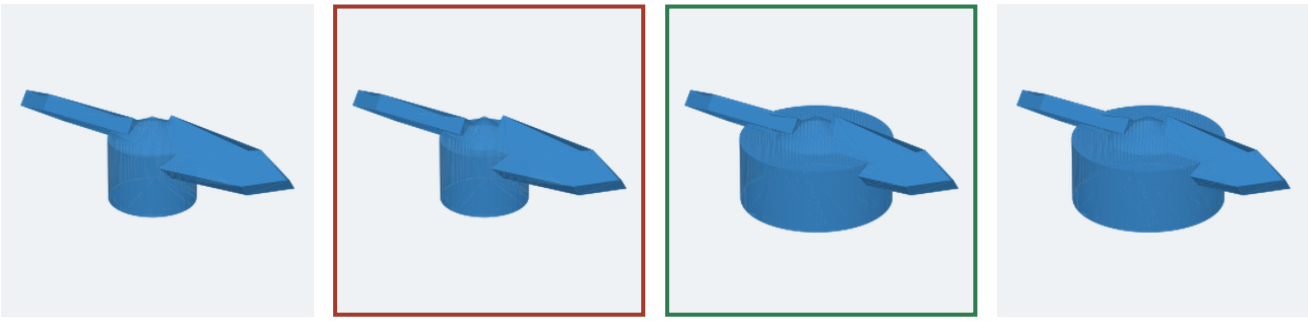
Reference
head radius 6.0 = 2 × shank 3.0

source, line 8	`.sphere(3.2)`
Qwen single	`.sphere(40.0)`
Qwen + pipeline	`.sphere(6.0)`

Caught by the semantics stage on rendered views: "The edited part shows a large sphere with no visible shank, violating the instruction that the shank should remain unchanged." One repair step gave the exact answer.

FIXED BY THE PIPELINE **Propeller: hub radius** `BenchCAD · tier T3 · aea612b8`

> *"Set the hub radius equal to half the blade length; the blades, spinner, and hub height stay the same."*



Source
hub radius 18.8

Qwen, single answer
IoU 0.442 · edited the spinner, not the hub

Qwen + pipeline
exact match

Reference
hub radius 32.3

source, line 3	`.cylinder(27.9, 18.8)  # hub`
Qwen single, line 5	`.circle(16.92) → .circle(12.5)  # spinner base`
Qwen + pipeline	`.cylinder(27.9, 32.3)`

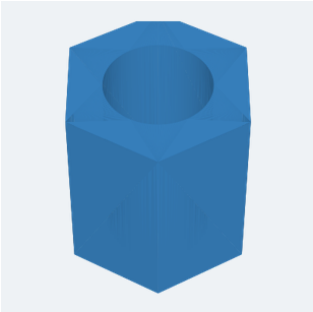
Caught by the semantics stage: "The edited part appears visually identical to the original part, showing no increase in the hub radius." The repair moved the edit to the hub line.

FIXED BY THE PIPELINE

Hex standoff: spherical bowl

BenchCAD · tier T5 · 677b012f

> *"On the top, carve a spherical bowl of 5.5 mm radius centered on the central axis."*

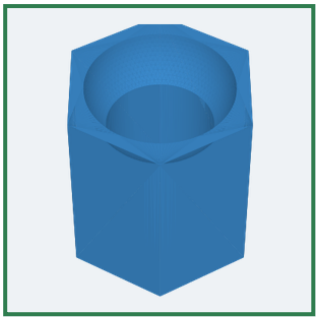


Source
hex prism with an 8 mm bore

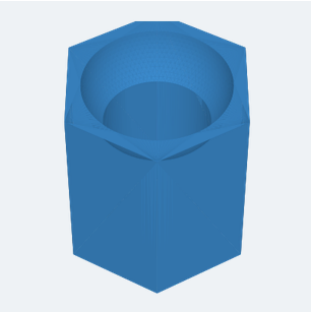
did not build

TypeError: Workplane.cut() missing 1 required positional argument: 'toCut'

Qwen, single answer
crashed; also deleted the bore



Qwen + pipeline
exact match



Reference
bowl cut at the top centre

Qwen single, line 6	`.hole(8.0) → .sphere(5.5).cut()`
Qwen + pipeline	`bowl = cq.Workplane("XY").sphere(5.5).translate((0, 0, 15.0))`
	`result = result.cut(bowl)`

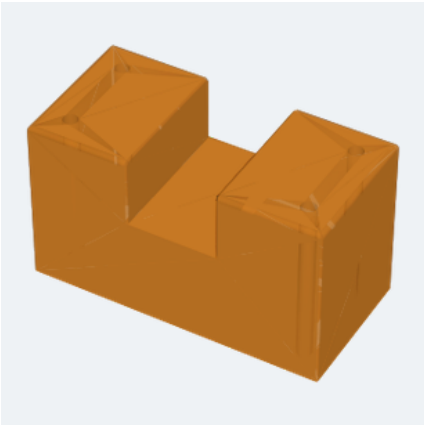
Caught by the geometry stage: the script did not run. The diagnosis passed the traceback to the planner, and the first repair passed every stage.

ASTRA MISS · PASSES EVERY CHECK

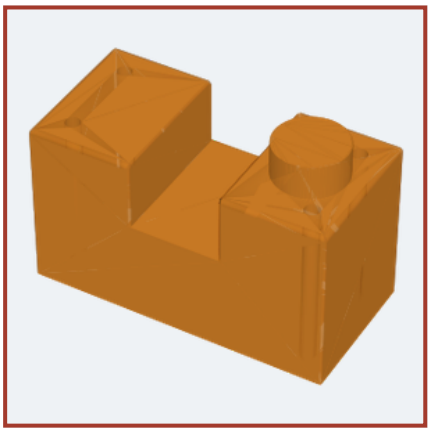
Locator block: centred boss

BenchCAD · placement error · 36fbd041

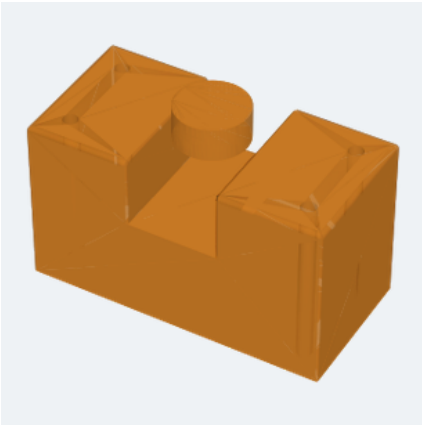
> *"Add a raised round boss centered on the top face of the locator block, 30 mm in diameter and 14 mm tall, sitting flush above the existing top and flanking the rectangular pocket."*



Source
block with a central pocket



Astra
IoU 0.959 · boss on the right-hand block



Reference
boss at the part centre, over the pocket

Astra	`.faces(">Z").faces(">X").workplane(centerOption="CenterOfMass")`
	`.circle(15.0).extrude(14.0)`
Reference	`cq.Workplane('XY').transformed(offset=(0, 0, 42.75)).cylinder(14.0, 15.0)`

Why nothing catches it: the diameter, height and "solid changed" checks all pass, and the rendered views look like a reasonable boss. This is one of the 13 genuine Astra errors that pass every target-free check, which is why checkers and repair loops can't improve Astra's 89. Qwen also misses this task, with and without the pipeline.

5. Five attempts to improve Astra, none beat 89

Trishita's suggestion: keep Astra as the draft model and add a trained checker on top. Astra is the only draft model; Claude was set aside. Every variant was evaluated on Astra's saved drafts for the 124 test tasks, so no new API calls were needed.

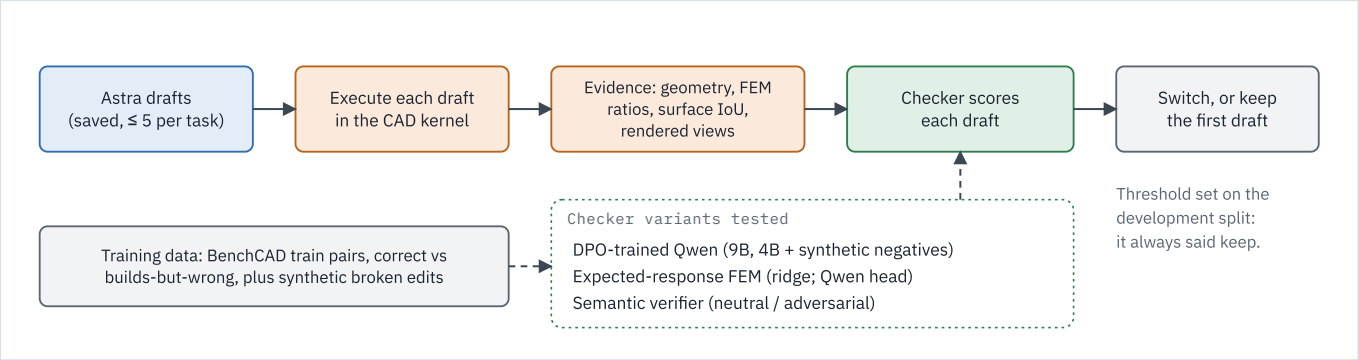


Figure 3. Checkers on top of Astra. Astra stays the only draft model. Each checker scores Astra's saved drafts from execution evidence; a threshold set on the development split decides whether to switch away from the first draft.

METHOD	WHAT IT LEARNS	PICKS THE RIGHT DRAFT	FINAL / 124
DPO checker, Qwen3.5-9B LoRA (H100)	Correct vs builds-but-wrong drafts; 213 pairs, plus geometry, FEM and surface IoU evidence	66–68% untrained 58%	89
DPO checker, Qwen3.5-4B + synthetic negatives (Kaggle)	1,379 pairs with synthetic broken edits; single-token DPO loss	71% held-out pairs 77.7%	89
Learned FEM score (linear)	Whether a draft's change in stress and compliance looks right	56%	89
Expected-response FEM (ridge)	Predicts the correct edit's compliance and stress ratios; $R^2 = 0.39$	68% ceiling 92%	89
Expected-response FEM (Qwen3.5-4B head)	Same targets with an LLM; failed to learn (R^2 0.09 / −0.52)	50%	89

In every case the development split says "never switch away from Astra's first draft", so the final score stays 89. Forcing switches gives 85–89. The FEM checker is a continuous reward (predicted response), not a binary pass/fail.

Semantic verifier and the two-player repair game

A repair game (repairer vs adversarial verifier) can only help if the verifier catches Astra's real errors without flagging its correct answers. We tested that first: Qwen3.5-9B read features extracted from each built CAD model plus the instruction, on the 120 Astra answers that build.

VERIFIER	REAL ERRORS FLAGGED	CORRECT ANSWERS FLAGGED	BENCHMARK ISSUES FLAGGED
Neutral prompt	12 / 19	26 / 88	6 / 13
Adversarial prompt	14 / 19	50 / 88	8 / 13
Needed to net a gain	most	≤ 3 / 88	–

Decision: no-go. A game driven by this verifier would break more correct answers than it fixes, so the full game was not run.

6. Why Astra is hard to improve: its 35 misses

Each of Astra's 35 BenchCAD misses was classified by hand (`docs/ASTRA_BENCHCAD_FAILURES.md`).



Benchmark issues (13)

- 6: reference contradicts the instruction
- 4: reference model is defective
- 3: instruction values are rounded

Astra follows the instruction in these cases, so they can't be fixed.

Genuine errors (22)

- 8: placement, frame or order of operations
- 6: edit scope (changes too much or too little)
- 5: derived geometry
- 3: invalid output

13 of these 22 pass every target-free check we have, so no check-driven loop can see them.

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7. New metrics from CAD-Editor and AgenticCADedit

We added the metrics from CAD-Editor and AgenticCADedit (arXiv 2609.29621), plus RealCADBench surface and solid voxel IoU (arXiv 2609.03773): validity, Chamfer distance, voxel IoU, DINOv2 isometric-view similarity, JSD and directional CLIP. Failures count as 0 in the means.

BENCHCAD (123 TASKS)	EXACT	VALID %	CHAMFER $\times 10^{-3}$ ↓	VOXEL IOU ↑	DINOV2 ↑	JSD ↓
Astra	88	97.6	1.45	0.948	0.957	0.031
Qwen3.5-35B single	53	63.4	5.53	0.598	0.609	0.056
Qwen3.5-35B + pipeline	66	78.9	3.48	0.750	0.762	0.040
Gemini Flash-Lite single	22	39.8	19.59	0.340	0.365	0.097
Gemini Flash-Lite + pipeline	35	53.7	12.92	0.475	0.505	0.069
3B patch adapter (paper base model)	25	53.7	12.16	0.417	0.485	0.080

- The pipeline improves every metric for both planners, not only strict match.
- Directional CLIP is near zero for every system (−0.001 to 0.011), so it doesn't separate them, and we don't report it as evidence.
- One BenchCAD reference fails to build, so this table has 123 tasks; that is why Astra shows 88 here instead of 89. On CAD-Editor, 7 references fail and are excluded.
- Not done yet: the VLM instruction and quality ratings, which need a vision judge.

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8. Parametric CAD Bench v2 and v3 as new datasets

These are cadbench.ai's FreeCAD tasks. The model writes a PartDesign script, and the official validator scores the saved model against a hidden reference.

VERSION	TASKS	LOCAL VERIFIER	CHECK
v2	100 create from text	validator 0.4.0, the official setup	Re-scores Astra's published answers at 0.847831; every task matches within 10^{-4}
v3	30 create, 30 create+edit, 40 drawing-to-CAD	validator 0.6.0, FreeCAD 1.1.0, OCP 7.9.3.1	Reference answers score 100.0 on all 100 tasks; wrong answers score 0.0 on all 9 tried

Pipeline for FreeCAD scripts

`scripts/cadbench_pipeline.py` writes a single answer and independent alternatives, runs each in FreeCAD 1.1, and applies the official validator's reference-free integrity gates. It repairs failing branches from their diagnosis, then selects among passing candidates by geometric agreement. Graders and reference models never leave the Mac; GPU hosts only see the public instructions and drawings.

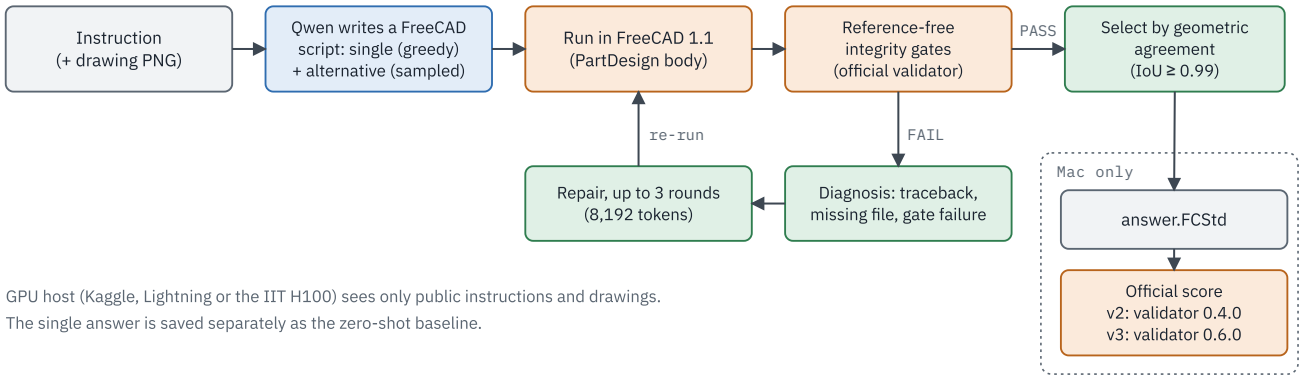


Figure 4. CAD Bench pipeline for FreeCAD scripts. The same plan, check, repair and select loop, with FreeCAD as the kernel and the official validator's reference-free gates as the checks. Official scoring runs only on the Mac.

RUN	MODEL	RESULT SO FAR	STATUS
Lightning Studio, H100	Qwen3.5-35B-A3B with thinking	First 13 v2 tasks: reference-free gates pass for 2/13 single answers vs 11/13 with the pipeline	STOPPED studio slept; not officially scored
Kaggle, 2×T4 (vLLM)	Qwen3.5-9B, no thinking	Round 0: 0/200 single answers pass. Round 1 repairs: 200/400 done at 18:08	RUNNING ends about 23:45 IST

The 9B model's first answers fail for repairable reasons: 130 scripts call `App.Placement(App.Vector(...))` without the required rotation, and about 100 hit the 4,096-token limit before saving. Repairs get 8,192 tokens. Official v2/v3 scores, single vs pipeline by task type, follow once the run ends.

9. Where Qwen fits (for Trishita's question)

SETUP	QWEN'S JOB	LEARNED FROM ASTRA?	ASTRA AT INFERENCE?	HELD-OUT RESULT
Distillation student (removed from paper)	Writes the edit	Yes, 441 Astra answers	No	68 / 252 Astra 106; a control trained without Astra data also 68
DPO checker	Judges Astra's drafts	No, correct vs wrong drafts	Yes, Astra drafts	89 / 124 same as Astra alone
Pipeline planner	Writes edits inside the pipeline	No, untrained	No	53 → 66 / 124

Qwen adapts at inference only through the pipeline: diagnosis, repair, search and an experience memory. The memory starts with the 492 BenchCAD training edits, and held-out tasks that pass every target-free check are added as the run goes (74 in the Qwen run). Weights never change during evaluation.

10. Compute and code

- **Kaggle 2×T4:** DPO checker v3, the FEM predictor, the semantic verifier and the CAD Bench run. vLLM 0.31 (CUDA 12.9 build) now serves Qwen3.5-9B in fp16 there after removing its broken `torchcodec` audio dependency. The plain transformers path ran about 110 s per answer.
- **Lightning Studio H100:** vLLM Qwen3.5-35B plus FreeCAD; used for the first CAD Bench run until the studio slept.
- **IIT H100 (trishita, 10.71.9.40):** used for the Qwen pipeline and the 9B DPO checker; reachable only from the IIT network.
- **Colab:** a self-contained CAD Bench notebook (`notebooks/CADBench_Pipeline_Colab.ipynb`).
- **GitHub:** commits `b525818` and `39fb71f7` are pushed. The Kaggle runner's vLLM and resume changes are not committed yet.

11. Open items

Finish the Kaggle CAD Bench run and score single vs pipeline with the official v2/v3 verifiers	TONIGHT
Finish repair rounds 2–3 in a second session if round 1 shows a gain	DEPENDS ON RESULT
Harbor login to download published v2/v3 runs (Astra, Claude) and score them with the new metrics	NEEDS LOGIN
VLM instruction and quality ratings (AgenticCADedit protocol)	NEEDS A VISION JUDGE
Live Astra pipeline run on all 124 tasks	NEEDS API CREDIT

Numbers come from run summaries under `cvpr2027/runs/`: adaptive-cad-open-20261007, adaptive-cad-gemini-lite-20261007, got-fem-astra-20261007, dpo-checker-benchcad-20261008, semantic-verifier-astra-20261008, astra-failure-analysis-20261008, edit-metrics-20261008, cadbench-v3 and cadbench-pipeline-kaggle-20261008. The Lightning CAD Bench figure is preliminary and has not been scored with the official verifier.